Problem B
Beacon Maintenance Network

Time limit: 3 seconds

A monitoring pipeline has n beacons in a line, numbered 1 through n. Each beacon is either **active** or **inactive**.

The engineer can service any active beacon. Servicing the beacon at position i does two things:

- Removes that beacon permanently, leaving a gap.
- Toggles each beacon at positions $i - 1$ and $i + 1$ that is still present. An active beacon becomes inactive, and an inactive beacon becomes active.

Inactive beacons cannot be serviced. Beacons keep their original positions: gaps never close, and an operation does not affect beacons across a gap.

The remaining beacons form **segments**. A segment is a maximal group of consecutive positions where no beacon has been removed. The pipeline is fully shut down when all beacons have been removed.

The pipeline also has an automated recovery system that can perform the same service operations. The engineer wants to reach a state where **every remaining segment contains exactly one active beacon**. From that point, later service operations cannot prevent a full shutdown.

Only the engineer acts during the operations used to reach this state. **The recovery system does not act during this sequence.**

The initial state is given by a binary string of length n, where 1 means active and 0 means inactive. Initially, no beacons have been removed.

An initial state is called **serviceable with budget f** if the engineer can reach the required state using **at most f operations**. Using zero operations is allowed. If no beacons remain, the required condition is also satisfied.

For each query (n, f), count the binary strings of length n that are serviceable with budget f.

Facts you may use

You may use the following facts without proving them:

- For an initially unbroken pipeline, **there exists an order of legal operations that removes all beacons if and only if the number of active beacons is odd**. This does not mean that every legal order succeeds.
- A segment with an even number of active beacons, including zero, can never be fully removed. Any service operation within such a segment leaves at least one segment with an even number of active beacons.
- If a segment contains more than one active beacon, the recovery system can choose a single service operation that leaves a segment with an even number of active beacons. Full shutdown then becomes impossible.
- If every remaining segment contains exactly one active beacon, any legal service operation leaves the same condition true for all remaining segments. Therefore, continuing to service active beacons will remove the entire pipeline, regardless of which active beacon is chosen at each step.

Input

The first line contains an integer q ($1 \le q \le 100000$), the number of queries.

Each of the next q lines contains two integers n and f ($1 \le n \le 1000000$, $0 \le f \le n$): the pipeline length and the maximum number of operations the engineer may use.

For each query, define

$$j_{\max} = \lfloor (n - 1) / 2 \rfloor, f_c = \min(f, j_{\max}),$$

and

$$w = \min(f_c + 1, j_{\max} - f_c + 1).$$

Queries may repeat. Across all distinct pairs (n, f_c) in one input file, the sum of w is at most $5 \cdot 10^7$. Count each distinct pair only once when calculating this sum.

Output

Print q integers in query order.

For each query, print the number of serviceable binary strings of length n, modulo 1000000007.

Sample Input 1	Sample Output 1
4 3 0 3 1 3 2 3 3	3 4 4 4

Sample Input 2	Sample Output 2
5 4 0 4 1 4 2 4 3 4 4	4 8 8 8 8